# Content-based Image Retrieval

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***Abstract*—A picture is worth a thousand words. Yes, but which ones? Content-based image retrieval (CBIR) is the application of computer vision to the image retrieval problem. The image retrieval problem is the problem of searching for digital images in large databases. "Content-based" means that the search will analyze the actual content of the image. The term 'content' in this context might refer to colors, shapes, textures, or any other information that can be derived from the image itself. Without the ability to examine image content, search must rely on metadata such as captions or keywords, which may be laborious or expensive to produce. How this problem is used in real world examples? Can algorithms really guess the similarity which is subjective category? Can similarity works without the context? How this reflects the consumer approach to online services? The presented paper gives some ideas that should clarify the raised questions.**

***Keywords*—CBIR, Metadata**

## I. INTRODUCTION

A picture is worth a thousand words [1]. It is a proverb that refers to the idea that complex stories can be described with just a single still image, or that an image may be more influential than a substantial amount of text. On Fig. 1 is a graph of sine function that tells us much more about function itself than the whole logarithmic tables.

Currently there are two general approaches to the storage and retrieval problems of digital images. One of these is the use of manually entered textual image descriptions. In this approach, descriptive text is added to images as metadata in order to be able to retrieve the images by querying for the added textual metadata. This added data could be technical information like time and date of creation, resolution, focal distance, etc., or some semantic description about the content of the image, like 'flowers', 'Conference' and 'Town'. Next to describing tactile objects or subjects in an image, more abstract or emotional concepts like 'colorful' and 'impressive' can be used as well. The use of text to describe the contents of images is also done because a computer is better at comparing text rather than images. Actually, it means that it is relatively easy to develop, because knowledge regarding text comparison has already been developed and text has a structure that can be used as a handle by the computer. In addition, manually entered descriptive texts are high-level features. When a description of an image is rich enough, the image is very well possible to turn up from a database after entering a matching query. However, manually adding textual metadata to all newly created images is a time consuming and tedious process.

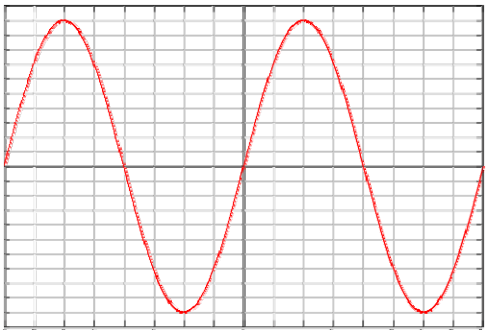


Figure 1. Sine function

It is sensitive to type errors and limited by the vocabulary of, and affected by the (cultural, historical or educational) background of the person that enters the data. One person might call something a dot, while another calls the same shape a circle. One person sees orange, while another calls the same color yellow.

The second general approach to image storage and retrieval is indexing images by computer vision. In this approach, images submitted to the system are analyzed by the computer. The computer then selects features from a wide spectrum of optional features like color, shape or structure, which best suit the automated description of the image. These directly visually derivative features are low-level features. They are measurable. The low-level feature-description, the collection of visual characteristics of an image is stored in a database. Images are retrievable from the system by looking for the stored low-level image features later on. This approach is known as Content Based Image Retrieval (CBIR).

## II. KEYWORDS – DESCRIPTIVE METADATA

Manually entered descriptive metadata (MEDM) is a chosen term for the process of describing the contents of an image with text. It is a method for categorizing images.

The descriptive text regularly has the form of separate words, which are known as 'keywords' or 'tags'. This way of indexing makes it possible to place images in a context. As a result, images can be retrieved by looking for the textual references to, and thus contexts of the images.

Search-by-keyword systems may suffice in serving the user when a searched image is a specific picture about a certain subject, event, object or person. However, as said before, manually entering descriptive data is not a solution people like to use. It is merely useful for a small selection of images, which the creator of the image wants to get to other's attention. The Flickr website (www.flickr.com) for example, is a very popular site for exposing and sharing

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images that utilizes keywords for indexing them. The site gives users the opportunity to upload imagery and add keywords that they think are relevant in order to find the images with a keyword search.

MEDM is quite usable for image databases where the domain of the image collection is narrow and people, who have the time and experience for this, do the filtering of the images. For describing medical images, one requires different knowledge and experience than comparing company logos for copyright-related issues, which requires special training as well. With a narrow domain comes a defined language space, or jargon. Users of systems within a domain know of conventions and textual terms, which relate to the system's context.

When the image retrieval system needs to store, index and retrieve images from a broad range of subjects, issues arise. These issues come forth from interpretations that are influenced by personal, cultural, educational and psychological experience.

### *A. Personal interpretation*

A mood or state of mind can influence the way an individual interprets text or images. A person plays different roles in different locations and in different social circles. With this way of behaving comes a mental state. At home, one uses a different jargon than in a professional environment. A single individual can choose different words to store or search for the same subject, depending on where she is [2]. The search task or goal can even change during an exploratory search for images, when the images that are returned by the system inspire the user to think differently.

Every other individual has her own interpretations. One person may use the word 'circle', as the other might choose for 'ball' and yet another for 'sphere'. When photos are tagged for personal use, this is not so much of an issue. There is a high chance that a user chooses the same word as index tag for a photo as she would for retrieving that photo from the system. In this case, she might even relate the tag directly to that photo only - not to other images with a 'circle' in it.

### *B. Cultural infulences*

What someone thinks is important, relies highly on the cultural background of that person. The environments in which people live affect their thoughts, norms, values, aesthetic valuation, and consequently interpretations and name conventions.

### *C. Language*

Language is also an issue with MEDM-based image databases. Before the development or first use of a tagged-based image database, the system's main user language needs to be determined. This way, every user is expected to use the same language, which is less a problem when the system is used in a single country.

Tag-based image databases are very useful for the internet. This is mainly because searching in and selecting from textual databases is fast and compatible with web-based databases. Online image databases are available to many countries since internet is intrinsically accessible from all over the world.

English is the main used language online. Still, there are many countries of which the level of English skills is not high. This makes English based systems hard to use to non-English native speakers. Misspelled keywords for indexing photos (because of a type-error or lack of language skills) result in erroneous image retrieval.

### *D. Example*

Considering the previous discussion, sometimes keywords are not enough. Fig. 2 shows different retrieval results using the keyword 'tiger'.

As we can see, there are many different images with keyword tiger. The solution to find similar images to the desired one would be to use more keywords to narrow the search, or to use CBIR to find visually similar images to the requested one.

## III. CBIR SOFTWARE SYSTEMS AND TECHNIQUES

Content-based image retrieval, a sub-domain of computer vision, is a system in which a computer analyses an image to extract visual features. These features are known as low- level features. This means that the features are visually detectable, without assigning any semantic meaning to them. With CBIR, images can be indexed relatively quickly and easily. The user uploads an image and the system indexes it automatically. However, the many image feature extractions in CBIR are hard to develop in comparison to MEDM-based systems. The features are also hard to understand for end users. In addition, lack of semantics gives bad results after querying for an image from a CBIR system. This lack of semantics in CBIR-based systems is known as a semantic gap.

Different implementations of CBIR make use of different types of user queries.

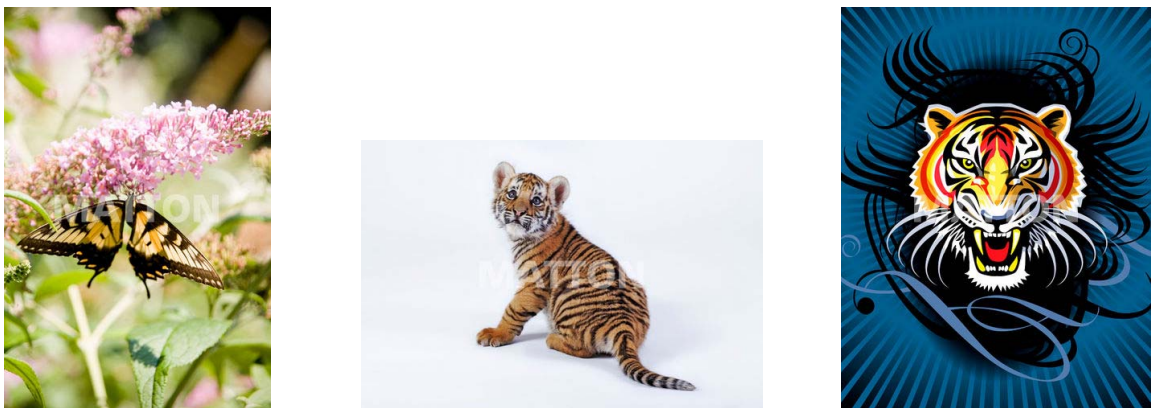


Figure 2. Different retrieval results using the keyword 'tiger' [3]

### A. Query by example

Query by example is a query technique that involves providing the CBIR system with an example image that it will then base its search upon. The underlying search algorithms may vary depending on the application, but result images should all share common elements with the provided example.

Options for providing example images to the system may include a preexisting image that may be supplied by the user or chosen from a random set, or provide a possibility for the user to draw a rough approximation of the image they are looking for, for example with blobs of color or general shapes.

This query technique removes the difficulties that can arise when trying to describe images with words.

### B. Semantic retrieval

The ideal CBIR system from a user perspective would involve what is referred to as semantic retrieval, where the user makes a request like "find pictures of dogs" or even "find pictures of Mickey Mouse". This type of open-ended task is very difficult for computers to perform - pictures of Chihuahuas and Great Danes look very different, and Mickey Mouse may not always be facing the camera or in the same pose.

People are able to do something else that computers cannot, which is a great issue in the context of CBIR. They are able to see one depiction and recognize transformed versions of that same object. This is the Invariance-property of the Gestalt theory. In addition, they can imagine what an object looks like on the side that is not within view.

Next to that, they can translate certain characteristics from one object to another, enabling them to categorize the object the very first time they see it. When a person sees an Old English sheepdog, it is recognized as a dog. When the same person sees a photo Chihuahua, she identifies it as dog as well, regardless of the fact that it has a different color, shape and absolute size. The orientation and pose of both animals do not even need to be similar to be able to identify both as dogs. A cat however, features similarities in comparison to a dog. It also has four legs, a tail and two ears, yet people recognize it as a cat, not a dog.

### C. Other query methods

Other query methods include navigating customized/hierarchical categories, querying by image region (rather than the entire image), querying by multiple example images, querying by visual sketch, querying by direct specification of image features, and multimodal queries (e.g. combining touch, voice, etc.).

CBIR systems can also make use of relevance feedback, where the user progressively refines the search results by marking images in the results as "relevant", "not relevant", or "neutral" to the search query, then repeating the search with the new information.

### D. Content comparison techniques

The best-known examples of extractable image features are color, shape and texture. In case of color, we can think of the overall color of an image, or the color of separate shapes within the image. Features are extracted from every new image that is introduced to the system, by means of mathematical algorithms. The extracted image features are then stored in a database for each image. Storing as many as possible relevant features increases the chance of an image to be found. A user can then retrieve images from the system by looking for image features.

The subsections below describe common methods for extracting content from images so that they can be easily compared.

#### 1) Color

The world is not colored; we just see it that way.

Retrieving images based on color similarity is achieved by computing a color histogram for each image that identifies the proportion of pixels within an image holding specific values (that humans express as colors). Current research is attempting to segment color proportion by region and by spatial relationship among several color regions. Examining images based on the colors they contain is one of the most widely used techniques because it does not depend on image size or orientation. Color searches will usually involve comparing color histograms, though this is not the only technique in practice.

#### 2) Texture

Texture measures look for visual patterns in images and how they are spatially defined. Textures are represented by texels which are then placed into a number of sets, depending on how many textures are detected in the image. These sets not only define the texture, but also where in the image the texture is located. Texture is a difficult concept to represent. The identification of specific textures in an image is achieved primarily by modeling texture as a two-dimensional gray level variation. The relative brightness of pairs of pixels is computed such that degree of contrast, regularity, coarseness and directionality may be estimated. However, the problem is in identifying patterns of co-pixel variation and associating them with particular classes of textures.

#### 3) Shape

Shape does not refer to the shape of an image but to the shape of a particular region that is being sought out. Shapes will often be determined first applying segmentation or edge detection to an image. In some cases accurate shape detection will require human intervention because methods like segmentation are very difficult to completely automate.

Some software producers are aggressively trying to push CBIR based applications into the filtering and law enforcement markets for the purpose of identifying and censoring images with skin-tones and shapes that could indicate the presence of nudity, with controversial results.

### E. The computer is wrong

The user defines visual criteria for the retrieval of images she is looking for; she queries. Querying in CBIR can be done in several ways of interacting with an image retrieval system, as described. The combination of all indexed features of a single image should create a unique description of that image. This single image should be retrieved by the system when a user defines an exactly matching feature-combination as a query.

Literally, an endless amount of optional low-level features could be extracted from images. When indexing a new image by means of CBIR, the computer has to decide which features are most useful and which are less relevant

for indexing and later retrieving an image. Some features are more important than others are. Features that seem important because of size, amount of extractable information and redundancy are marked as such. Features extracted from smaller parts in an image, which contain less information are treated accordingly.

As said earlier, all features from one image together form a link to that single image. Photos contain coherence between features. For example, after a deduction of dimensions the following clues can all tell us we are looking at 'sky':

- It generally sits in the upper half of an image;
- It can have a variable size or shape;
- It often has a blue color, sometimes purple, pink or orange;
- If blue, a yellow or orange circle could be seen within that blue shape;
- The circle itself (if it actually is the sun, not a ball) would never be in the lower half if the blue shape were in the upper half;
- The circle's position within the blue shape is variable.

The computer cannot detect this coherence between all variable features, since recognition is based on semantics. Even if we would teach the computer these variables, would it be able to understand the concept of 'sky' when it is reflected in water? What happens when the sky is the larger part in an image, but a small silhouette in the lower-left bottom is relevant to the user? It is impossible to select the most important feature automatically, since the most important feature is never the same. This is not just the case at the indexing stage of images, but also at the time of retrieval. The importance of a feature depends on the user's task or goal and her personal interpretation.

The user will have to tell the system what is important, what is similar and what is different to her, to make it possible for the computer to retrieve what a user is looking for. Interaction is the key to satisfying image retrieval results.

### *F. People are wrong*

People can miss-interpret colors from an image. When the computer-measured color of a wall is grey, a person can interpret the wall as white.

Comparison by people is done differently than comparison by a computer. Like with a personal vocabulary, the amount of detail a user can detect depends on the personal experience with a sub-domain of images. An antiquarian for example, can see whether an old looking piece of furniture is actually antique, just old furniture or a new piece of furniture that is forged to look like antique.

One image in either JPEG format or GIF file-format looks the same to people, but very different to a computer, since they are two different file-types. The content of both the images is the same; however the images are actually quite different (mainly in color). People need to program applications in such a way that the computer sees these images as equal. In the field of CBIR, these problems could be solved. What if the same image is in black-and-white? At one time the color is important, another time the image is.

In Fig 3 we have two almost identical images to our eyes, but they are very different to computer. The only difference is different lighting setup in digital camera. CBIR algorithm will consider these two images very different, although they are not. People have built-in lighting invariance. Computer should "learn" the lighting invariance to consider these images similar.

## IV. CASE STUDY: STOCK PHOTOGRAPHY

Stock photography consists of existing photographs that can be licensed for specific uses. Publishers, advertising agencies, graphic artists, and others use stock photography to fulfill the needs of their creative assignments.

A customer who uses stock photography instead of hiring a photographer can save time and money, but can also sacrifice creative control. Stock images can be presented in searchable online databases, purchased online, and usually delivered via download or email.

There are obvious reasons why to use CBIR methods in stock photography. Customer should find the best image for his creative workflow.

The problems with implementing CBIR search in stock photography are the following:

- Huge amount of photography (millions of them);
- Fast response to a query is needed;
- Huge quality differences between underlying metadata.

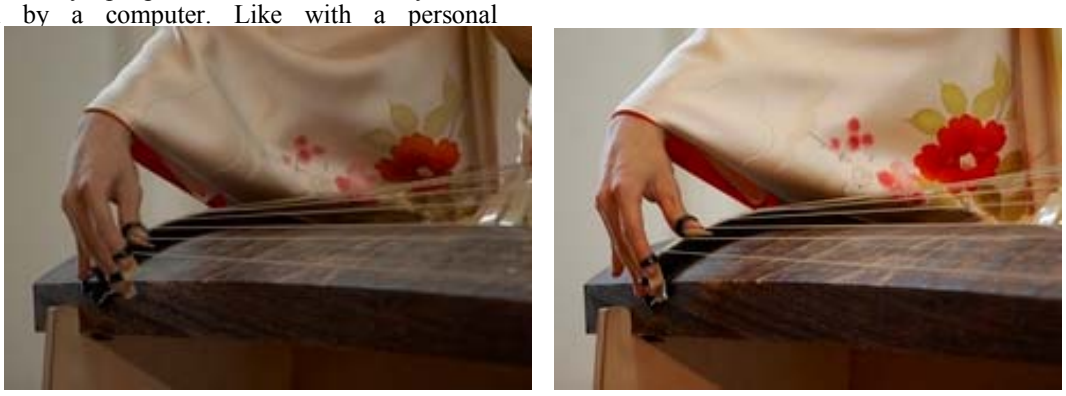


Figure 3. Different lighting setup in digital camera [3]



Figure 4. Finding similar images with the combination of usual keyword search and CBIR [3]



Figure 5. Finding similar images without putting them into some context [3]

The advantages with stock photography are the following:

- High quality of photography;
- Lighting invariance is not a problem, because images are normalized and post-processed to achieve the desired quality;
- A non-perfect result is actually an advantage. Even if the result is not showing the most similar (almost identical) images, customers would be satisfied, because the goal is to suggest alternatives to the reference image.

In Fig. 4 we can see how CBIR can help finding similar images. The best method, which was used here, is to combine usual keyword search with CBIR to get the most desirable results.

Without putting images into some context, we can get images which are very different to human, because human usually connect images with the subject of the image (Fig. 5).

However, sometimes only the color and texture – both of them make the mood of the image – are enough for finding similar images (Fig. 6 and 7).

### *A. Consumer feedback*

The regular consumer is rather a browser than a searcher. However, nowadays the consumer is also evolved from just a consumer of a product, to a partner in a process of adding value [4]. Therefore consumer pleasure should also be considered in the process of finding the desired photography.

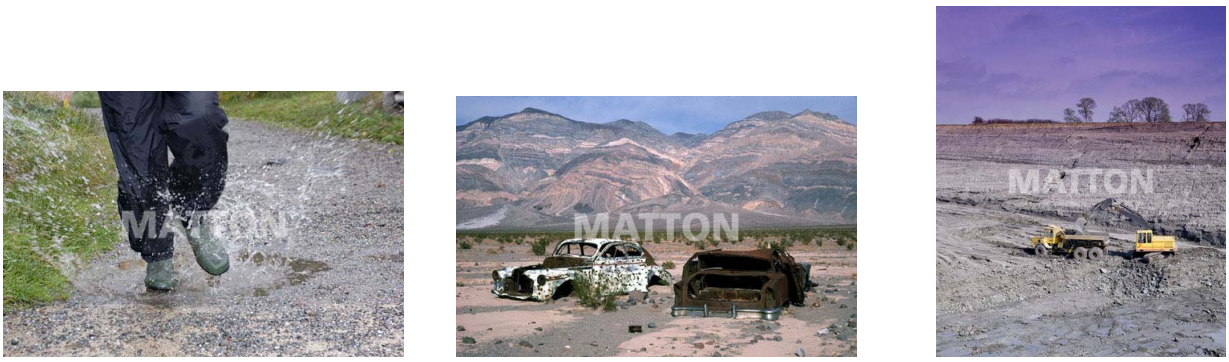


Figure 6. Color and texture make the mood of the image [3]



Figure 7. Color and texture make the mood of the image [3]

The literature describes a spectrum of research in this area and has identified a number of design considerations, which include:

- Minimizing the potential complexity experienced by the customer, keeping their expenditure in the querying process as low as possible, whilst providing clearly perceptible benefits;
- Reducing cognitive overhead, which lies not only in extent of choice, but also in areas such as lack of understanding about which solution meets their needs, uncertainty about the behavior of the supplier, etc.

One of the works that explicitly address the creation of positive feelings in product use is particularly useful as a hypothesis generating tool since it provides not a theory of pleasure, but rather a tool to help those involved in the design process take a structured approach to understanding the entire spectrum of pleasures a product can bring [5]. It identifies four types of pleasure associated with products:

- Physio-pleasure – relates to the body and is concerned with positive feedback from the sensory organs; touch, taste, smell, hearing and sensual pleasure;
- Socio-pleasure – refers to relationships with others – individuals, groups and society as a whole. Socio-pleasure is drawn from the aspects of products that confer social, material or cultural status, help to construct personal identity and/or stimulate desirable social interaction. These product qualities give positive feedback to the owners about their personal view of themselves in society;
- Psycho-pleasure – refers to a user's cognitive interaction with a product and their subsequent emotional reaction. Psycho-pleasure is drawn from products that give emotionally satisfying results from the cognitive demands of interaction;
- Ideo-pleasure – relates to peoples' values. In the context of products this ranges from aesthetics to ethics. It includes taste, moral values and personal aspirations. It defines how people do, and would like to, see themselves.

In case of stock photography, several types of pleasure are considered:

- Huge amount of photography, as well as high quality of them results in satisfactory Physio-pleasure and also Socio-pleasure;
- Easy querying and fast response to a query, as well as a non-perfect result lead to satisfactory Psycho-pleasure;
- The content of photography used results in Ideo-pleasure.

## V. CONCLUSION

The results of the case study show that the best approach is to combine two methods.

Without putting images into some context, we can get images which are very different to human, because human usually connect images with the subject of the image.

Even if we use only CBIR method without underlying metadata, some of the consumers can get satisfying results, because they need images which has similar mood**,** and very often color and texture can provide mood.

If picture is worth a thousand words, is that true that 1001 words worth more than a picture? We have more and more web services, especially social networks, where we get user generated context, instead of context generated by owner of the service itself.

Very often, tagging of data (especially tagging of images) is done by many members of the social network, and the user generated context which is displayed to the user is highly personalized to that user. Is this 'new wave' of information enough to help tagging of data, especially images, or we will still rely on CBIR?

The truth is probably between these two opposites.

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